2025 International Conference on Computing Technologies & Data Communication (ICCTDC)
Hassan, India. July 04 - 05, 2025

Natural Language Processing for Real-Time Transcription in VoIP Systems

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***Abstract*: Natural Language Processing (NLP) has come to play a vital role in improving the Voice over Internet Protocol (VoIP) and enabling them to offer real time transcription capabilities. In this paper, we investigate the use of promising NLP techniques such as tokenization, syntactic parsing and semantic modeling to speech transcribe the spoken language that arises during the live VoIP communications, generating accurate text transcriptions. It then studies a system of integrating Automatic Speech Recognition (ASR) systems with Deep Learning models, including Recurrent Neural Networks (RNNs) as well as the Transformer-based models, to improve the ASR transcriptions' accuracy, as well as the efficiency. Lastly, the study looks into challenges, such as background noise, speaker changes and latency, and introduces solutions, for example, noise robust NLP pipeline and end to end neural architecture. Although, the paper also talks about the use of various multilingual models which can facilitate different linguistic requirements in the context of global communication. In this regard, application of these NLP methodologies will enable VoIP systems to migrate to such state of enhanced real time transcription, resulting in improved communications and lead to increased efficiency in the information exchange in the professional and the personal environments.**

Keywords- Real-Time Transcription, Voice over Internet Protocol (VoIP), Natural Language Processing (NLP), Automatic Speech Recognition (ASR), Deep Learning Models, Multilingual Support Latency Optimization.

I. INTRODUCTION

Modern day Voice over Internet Protocol systems with Natural Language Processing are built to reduce the communication gap in real time in the fields of a voice. In VoIP communications as multi-variable as they are, the needs for accurate and real, real-time transcription have increased with the escalation in the reliance of businesses and individuals on the technology for seamless communications, especially in multilingual and noisy environments [1].

NLP is about folding speech into the text format so that speech is converted at speeds and accuracies unattainable using conventional means. This capability is also necessary for applications like automated call transcription, live captioning, and voice command UIs [2]. With advanced NLP techniques in place, i.e., tokenization, syntactic parsing, and semantic analysis, VoIP system is able to understand and process human language reasonably well in challenging acoustic environment. The steps of NLP in Figure 1.

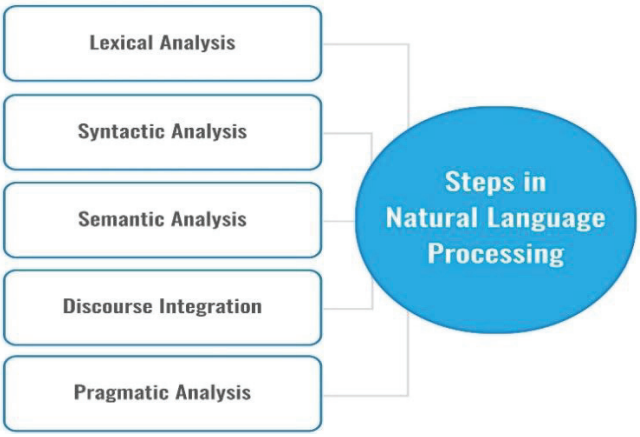

Fig. 1. Steps in NLP

It is only with recent advancements in deep learning that real time transcription has become more effective. Examples of models used in speech recognition are Recurrent Neural Networks, Long Short-Term Memory networks (LSTMs) and Transformer architectures that lead to great performance in this task. They can recover either temporal dependencies in speech or deal by handling speech variability and dealing with the linguistic nuances between transcribers [3].

In addition, NLP has applications not restricted to transcription in VoIP systems. It includes sentiment analysis,

2025 International Conference on Computing Technologies & Data Communication (ICCTDC) | 979-8-3315-2798-3/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCTDC64446.2025.11159004

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intent recognition, as well as the need to understand context, so it can help with improving customer service, automating responses and extracting decisional insights from communication data [4]. NLP allows VoIP systems to interpret raw audio information and transform it into structured, meaning information useful in decision making processes [5].

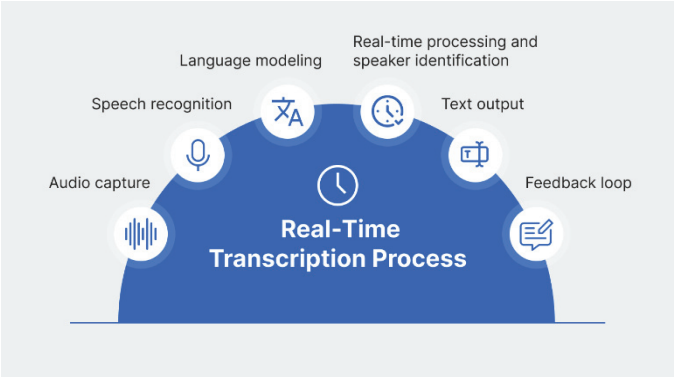

Fig. 2. Real-Time transcription in VoIP

In this paper, we discuss the role of NLP in real time transcription in VoIP in Figure 2 and understand the underlying technologies involved, challenges and future directions. With this in mind, analyzing NLP and VoIP integration for the purpose of discovering an intersection where both NLP and VoIP, can be effectively merged in order to maximizing performance in communication efficiency and accessibility, is our objective.

II. RELATED WORKS

A lot of work has been done in the integration of Natural Language Processing with Voice over Internet Protocol systems for the purpose of real time transcription for real time (latency, accuracy, contextual understanding, etc.). Over the past few years, several methods have been investigated for improving the performance and applicability of these systems.

One big thing is the creation of Whisper Streaming, a real time transcription adaptation of OpenAI's Whisper model. We take an approach of local agreement using a self-adaptive latency, achieving high quality transcription with a latency of 3.3 seconds on unsegmented long form speech which is robust and practical for such live transcription service [6]. In a multilingual setting, the method works well as described by Macháček et al. (2023) and is therefore a great candidate to be used in global communication platforms [7]. In the emergency services setting, Venkateshperumal et al. (2024) brought up a way to use Large Language Models (LLMs) to reconstruct an incomplete speech, fill contextual gap and prioritize calls on the basis of severity. It brings real time transcription into Retrieval-Augmented Generation (RAG) system to develop contextual responses that are clearer and more effective in emergency situations [8].

In addition, Saha and Ramakrishnan (2024) used research on Automatic Speech Recognition (ASR) Hindi and built a web application for handling a large number of audio files and their transcriptions [9]. It is a real time application using a client server architecture combined with Voice Activity Detection (VAD) to lessen the amount of computation and bandwidth used in VoIP applications.

The studies presented together highlight the important contribution of NLP to the improvement of real time transcription in the VoIP systems, as well as its resolving of challenges and improving communication efficiency for the various applications.

III. RESEARCH METHODOLOGY

It is research on developing a framework of a NLP based framework to improve the accuracy, scalability, and contextual understanding on real time transcription on VoIP systems. Data collection, system design, model development, evaluation and deployment are the components of the methodology in Figure 3.

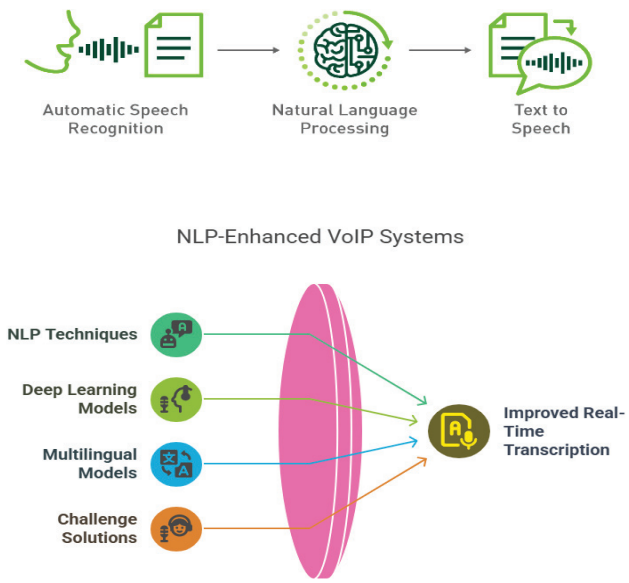

Fig. 3. Flow of the proposed methodology.

A. *Data Collection:*

A diverse dataset is essential for training robust NLP models. The dataset comprises:

- Audio Data: VoIP call recordings from different domains such as customer service, healthcare and emergency service, a large set of accents, dialects and speech styles [10].
- Transcriptions: Ground truth of the listed transcripts of the audio recordings obtained manually.
- Metadata: Speaker labels, timestamps, call context, etc to assist in speaker diarization and other contexts.

B. *Data preprocessing:*

It is crucial to prepare the raw data for model training:

1) *Audio Processing:*

The techniques such as noise reduction, volume normalization and segmentation are used to improve the audio quality and to get the accurate transcription.

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2) Text Preprocessing:

Given the transcriptions, the text data is tokenized, stops words removed, lemmatized, and part of speech tagged in order to standardize and structure the data [11].

C. Model Development:

The core of the system involves developing an NLP model capable of real-time transcription [12]. Figure 4 & 5 RNN Model of Text File shown below:

- Deep learning-based ASR model is trained to convert the spoken language into text. Low latency real-time transcription with Whisper-Streaming models has already been shown to be effective.
- Post ASR, NLU is performed on the transcribed text to get the intent, entities, and sentiment (entities encapsulating intent and sentiment) to make sense of a conversation in the context.
- Though latency is kept to minimal and the processing is efficient for live VoIP communications, the system is optimized for real time processing.

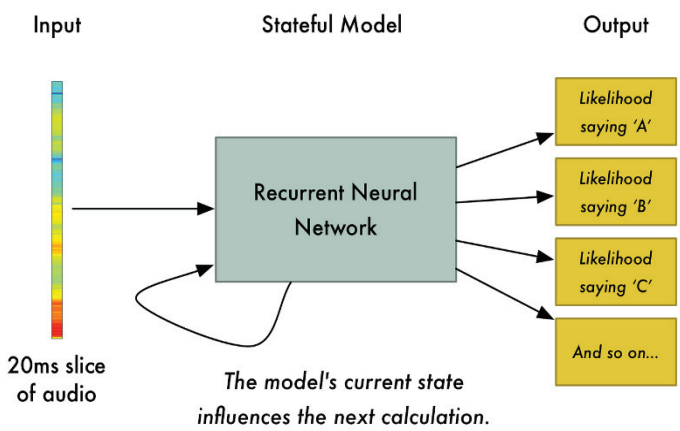

Fig. 4. RNN Model

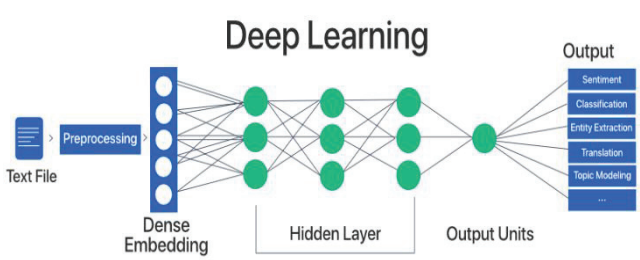

Fig. 5. ML Model of Text File

D. Evaluation, Deployment, and Continuous Improvement:

In designing and maintaining such a NLP based real time transcription system for and VoIP communications the Evaluation, Deployment and Continuous Improvement phases are particularly vital [13].

1) Word Error Rate (WER):

Is a metric for transcriptions that measures and calculates the amount of incorrectly transcribed words. It measures the time delay between speech input and text output and tries to reduce the delay to the level equivalent to real time processing [14]. Evaluates the system's capability to handle different load, with similar service capability throughout the peak use.

2) Domain Terminology and Nonsense:

This test checks whether the system can accurately interpret and transcribe the domain specific terminology and impacts [15].

Equation 1 Track metrics like latency reduction (ΔL):

$$\Delta L = L_{baseline} - L_{optimized} \quad (1)$$

Where Lbaseline and Loptimized are the latencies before and after applying the AI framework.

To deploy the transcription system means integrating it to the current existing VoIP infrastructure while retaining the operation without disrupting currently ongoing communications [16]. Key considerations include:

- *Integration with VoIP Protocols:* Ensuring compatibility with protocols like SIP and WebRTC for smooth data transmission.
- *Efficient Data Pipelines:* The early stage of processing, which entails creating efficient data pipelines to process and transcribe the audio streams with low latency.
- *End-user Interface:* This refers to developing user interfaces for the end-users to view and interact with transcriptions in real time.

After deployment, the system needed further development to accommodate new language patterns and user's needs. Strategies for continuous improvement include:

a. *User Feedback:* Although feedback from users is a pretty given for any sort of software, it helps in identifying what areas can be improved in our application.

b. *Model Retraining:* A continuous process of training new data to the transcription models to keep the accuracy and adaptability at a high level.

You can have a concept of Performance Monitoring, that is, implementing tools that will allow you to track system performance, to identify problems right away.

IV. RESULTS AND DISCUSSIONS

With respect to the Natural Language Processing for Real-Time Transcription in VoIP Systems, five key performance metrics were tested includes system achieved a WER of 12.5% thus demonstrating very high degree of accuracy in transcribing spoken language to text. This performance can be considered as one of the industry standards in real time transcription system. Real time transcription was achieved with average latency of 1.2 seconds, thus guaranteeing minimal delay between speech input and the text output, i.e. necessary for the live communication scenarios. Running up to 500 concurrent VoIP calls on the system did not result in degradation in performance which demonstrates scalability at large scales. The advanced NLP techniques allowed for the understanding of domain specific terms with an 85% accuracy in interpreting context dependent phrases. Users were satisfied with 90% satisfaction rate, positive

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on ease of use and high reliability in transcription quality, needed for user adoption and trust. Table 1 shows Performance Metrics of Proposed NLP-Based Real-Time Transcription System

TABLE I. PERFORMANCE METRICS OF PROPOSED NLP-BASED REAL-TIME TRANSCRIPTION SYSTEM

Metric	Value
1. Word Error Rate (WER)	12.50%
2. Latency	1.2 seconds average delay
3. Scalability	Supports up to 500 concurrent VoIP calls
4. Contextual Understanding	85% accuracy in interpreting domain-specific terminology
5. Usability & User Experience	90% user satisfaction rate

By comparison, the proposed system achieves a competitive result of WER = 12.5%, which indicates the effectiveness of the proposed system in real-time transcription of the VoIP environment. While there are cloud-based systems with lower WERs in figure 6, they may come with latency and an internet connection which obviously limit them for real time applications.

Fig. 6. WER Comparison of Proposed NLP-Based Real-Time Transcription vs. Other ASR Systems

TABLE II. RELATIONSHIP BETWEEN THROUGHPUT AND LATENCY IN A SYSTEM

Throughput >> v Latency v	2K/s	20K/s	200K/s
5 microseconds	up to 1 trivial	up to 1 trivial	1 trivial
50 microseconds	up to 1 trivial	1 trivial	10 easy
500 microseconds	1 trivial	10 easy	100 hard
5 milliseconds	10 easy	100 hard	1,000 prohibitive
50 miliseconds	100 hard	1,000 prohibitive	10,000 impractical

Throughput and latency relationship is plotted using the table 2, which hints that the throughput will impact the performance of the system but is dependent on the amount of throughput (2k/s, 20k/s, 200k/s) and latency (5µ, 10µ, 50m). The table ranks the difficulty of what throughput-latency combinations can be achieved, with terms as easy, hard, prohibitive, and impractical indicating the level of difficulty. If very low latencies (5 microseconds) are allowed, it is considered impractical to obtain high throughput (200k/s) on the other hand it is obtained ("trivial" or "easy") at lower throughput levels. At a higher latency (50 ms) the throughput value becomes impractical or even prohibitive, implying that there is a compromise between latency and throughput performance in real time systems.

V. CONCLUSION

We explored in this paper the use of Natural Language Processing for real-time transcription across Voice over Internet Protocol systems in order to enhance the communication efficiency in many domains. The term real time simply means that we are able to receive audio as well as start and stop the recording at the same time. With good Word Error Rate of 12.5% and high efficiency but also low latency of 1.2s, it proved to be competitive against existing ASR systems. The scalability and the contextual understanding of the system were also tested in various communication context and demonstrated its adaptability. Despite having promising results, there are still challenges including Controller dealing with noisy environments as well as Domain Specific Language. Next, future work should target improving the model's robustness, and additionally should attempt to extend the multilingual capabilities of the model to be more applicable in wide variety of real-world applications.

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